

gives us details like how many tests are included in this, as well as the hardening index of our operating system (as shown in Figure 1).

```
jishnu@kali: /etc/lynis
File Actions Edit View Help

Follow-up:
- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

Lynis security scan details:
Hardening index : 63 [##### ]
Tests performed : 259
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [7]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

Lynis 3.0.0
Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)
```

Figure 1: Basic output with hardening index

Adding your own script to Lynis

Let us now move on to the main aim of this article, i.e., to create one's own Lynis script. It indeed is very exciting to add one's own script to such a famous tool and is an easy procedure too.

First, we have to navigate to `/usr/share/lynis/include`. Here, you will find a file named `tests_custom.template`. This is a basic template in which one's script should be carried. This file is then copied with the same file name without the extensions template. We then move on to the custom tests section (as shown in Figure 2).

```
fi
#
#####
#
# Add a new section to the screen output
InsertSection "Custom tests - Other"
#
#####
#
# Test : CUST-0040
# Description : Our second test, with a prerequisite test
```

Figure 2: Custom tests section of Lynis tools

Add the script given here below the 'Insert section' line; in this case I have written the script to check if the `cron` job scheduler is up and running properly or not. The script is given below.

```
#
# Add a new section to the screen output
InsertSection "Custom tests - Other"
Register --test-no CUST-1313 --weight L --network NO
--description "test to check if mysql is running"
STATUS="$(systemctl is-active cron.service)"
if [ "${STATUS}" = "active" ]; then
echo "The cron job scheduler service is running properly."
else
echo "The cron job scheduler service is not running."
fi
```

We then run the Lynis audit again using the command that we have already seen, `lynis audit system --quick` and get the output of the newly written test in the custom tests section (see Figure 3).

```
[+] Hardening
- Installed compiler(s) [ FOUND ]
- Installed malware scanner [ NOT FOUND ]

[+] Custom tests
- Start custom tests... [ OK ]
- Checking if everything is OK... [ NO ]
  This shows one level deeper [ WARNING ]
  And even deeper [ OK ]
OS: Linux [ OK ]

[+] Custom tests - Other
The cron job scheduler service is running properly. [ WARNING ]
- Checking if everything is OK... [ WARNING ]

[+] Plugins (phase 2)

-[ Lynis 3.0.0 Results ]-
```

Figure 3: Output of the new test that was included

So, from this output we can infer that the cron job scheduler service is working properly.

Do give it a try.

If you want to learn more about the security of your system, check out the official documentation of Lynis. You can have great fun learning about the vulnerabilities and the security of a system. The deeper we dig in, the more fun it is. The main aim of this article is to show you how easy and exciting information and cyber security auditing can be. **END** 🐼

By: Jishnu Saurav Mittapalli

The author is a full stack Web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.